

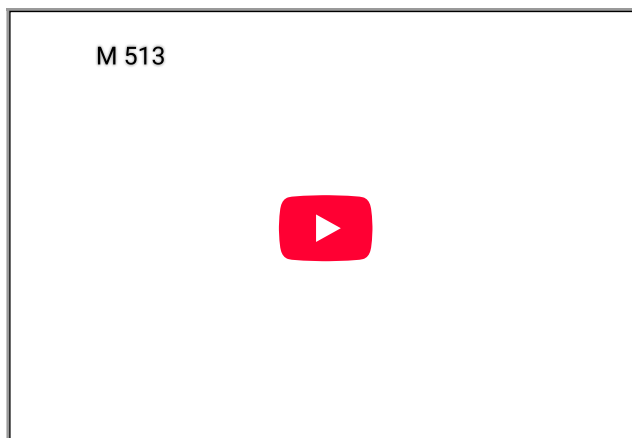
M 513: Instrumenting with Frida (15 pts extra)

Windows Users

If you are using a Windows system, do the [Windows version](#) of this project instead.

What You Need for This Project

- An computer with Android Studio installed



Purpose

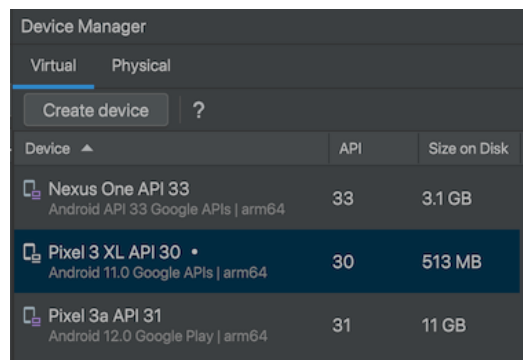
To practice instrumenting, overriding Java functions.

Launch an Android Emulator

In Android Studio, use an emulator with API 30 and a phone that does not have Google Play, so it's rooted.

Note: the app we're using doesn't work on API 33.

I used the emulator highlighted in the image below. My emulator uses an ARM64 processor. Make a note of the processor your emulator is using.



Installing Frida-Tools

On your host system, in a Terminal or Command Prompt, execute this command:

```
python3 -m pip install frida-tools
```

"error: externally-managed-environment"

If you see this error message, you need to create a Python virtual environment with these commands:

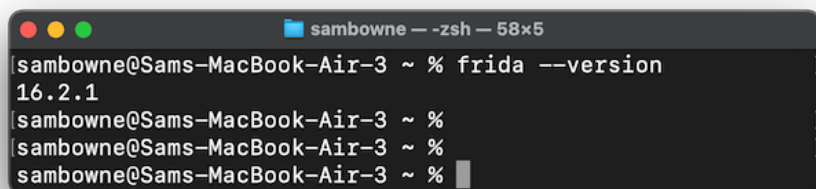
```
python3 -m venv venv
source venv/bin/activate
python3 -m pip install frida-tools
```

Execute this command:

```
frida --version
```

If it fails, try opening a new Terminal window and executing it there.

Note your version of Frida. When I did it in 2024, it was version 16.2.1, as shown below.



```
sambowne@Sams-MacBook-Air-3 ~ % frida --version
16.2.1
sambowne@Sams-MacBook-Air-3 ~ %
sambowne@Sams-MacBook-Air-3 ~ %
sambowne@Sams-MacBook-Air-3 ~ %
```

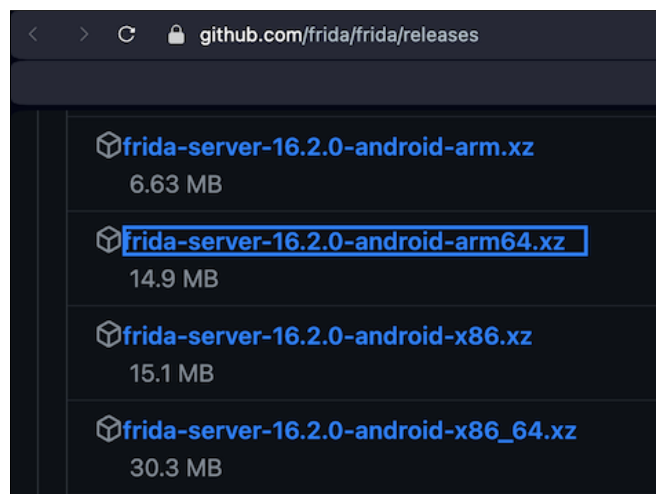
Downloading Frida-Server

On your host system, in a Web browser, go to

<https://github.com/frida/frida/releases>

Download the appropriate version of frida-server for your emulator's processor, with the same version number you found above, or as close as possible.

I used **frida-server-16.2.0-android-arm64.xz**, outlined in the image below.



Uncompressing Frida-Server

On a Mac, you can uncompress the file by double-clicking it. On Windows, use 7-zip.

Installing Frida-Server on your Android Device

On your host system, in a Terminal or Command Prompt, execute these commands:

```
adb push frida-server-16.2.0-android-arm64 /data/local/tmp/
adb shell
su
chmod 755 /data/local/tmp/*
/data/local/tmp/frida-server-16.2.0-android-arm64
```

The command hangs without replying, as shown below.

Leave this Terminal window open.

```
Downloads — adb shell — 94x5
^Cemulator_arm64:/ #
emulator_arm64:/ #
emulator_arm64:/ # chmod 755 /data/local/tmp/*
emulator_arm64:/ # /data/local/tmp/frida-server-16.2.0-android-arm64
```

Testing Frida-Server

On your host system, open a new Terminal or Command Prompt.

If your frida-tools were installed in a virtual environment, execute this command:

```
source venv/bin/activate
```

Execute this command

```
frida-ps -U
```

A long list of processes running on Android appears, as shown below.

```
128 — -bash — 65x14
Sam-2:128 sambowne$ frida-ps -U
PID  Name
-----
2087  adbd
2396  android.process.acore
2217  android.process.media
1396  audioserver
1397  cameraserver
3318  com.android.deskclock
1761  com.android.inputmethod.latin
1880  com.android.phone
2144  com.android.printspooler
2497  com.android.providers.calendar
1775  com.android.systemui
```

Installing a Vulnerable App

We'll use an intentionally vulnerable app.

On your host system, in a Web browser, go to

https://github.com/OWASP/owasp-mastg/blob/master/Crackmes/Android/Level_01/UnCrackable-Level1.apk

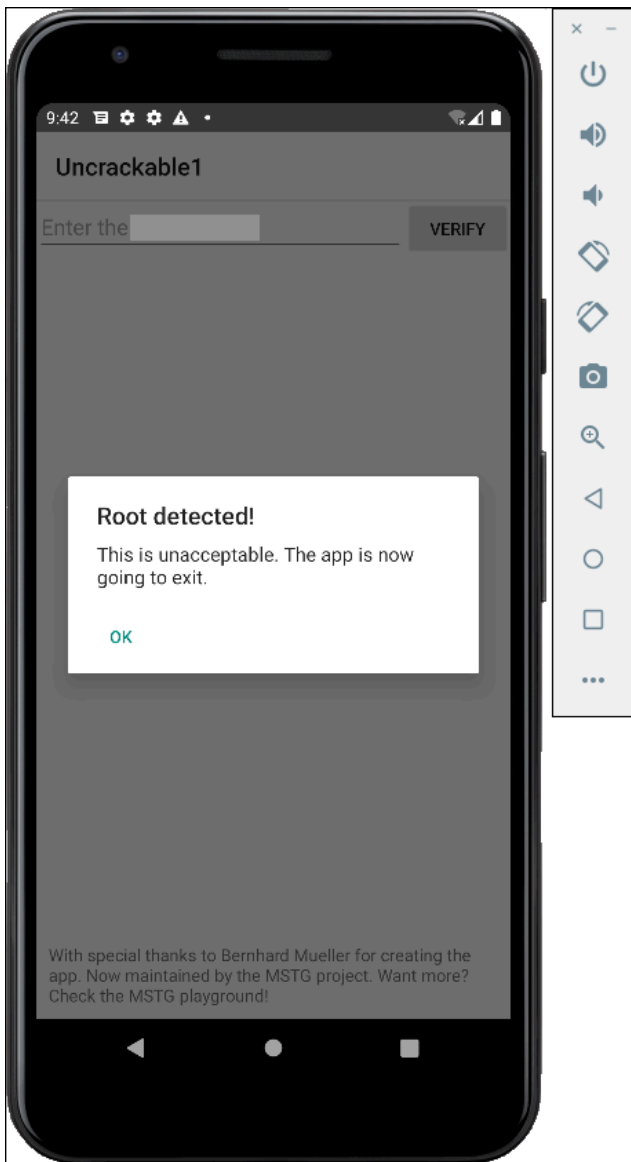
Download **UnCrackable-Level1.apk**.

If that page is not available, use this [alternate link](#).

Drag the **UnCrackable-Level1.apk** file and drop it on your running Android emulator.

On your emulator, launch the Uncrackable app.

It refuses to run, complaining about the phone being rooted, as shown below.



Viewing the Code in jadx-gui

Open the **UnCrackable-Level1.apk** file in jadx-gui.

Navigate to the Main Activity and click **OnCreate(Bundle)**, as shown below.

The root detection uses three functions: `c.a()`, `c.b()`, and `c.c()`.

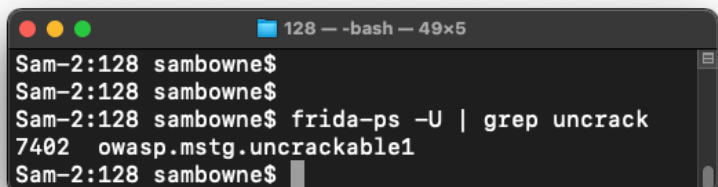
Finding the App's Name

On your host system, in an unused Terminal, execute this command:

```
frida-ps -U | grep uncrack
```

The name of the vulnerable app appears, as shown below.

The name is **owasp.mstg.uncrackable1**



```
Sam-2:128 sambowne$  
Sam-2:128 sambowne$  
Sam-2:128 sambowne$ frida-ps -U | grep uncrack  
7402 owasp.mstg.uncrackable1  
Sam-2:128 sambowne$
```

Creating a JavaScript Payload

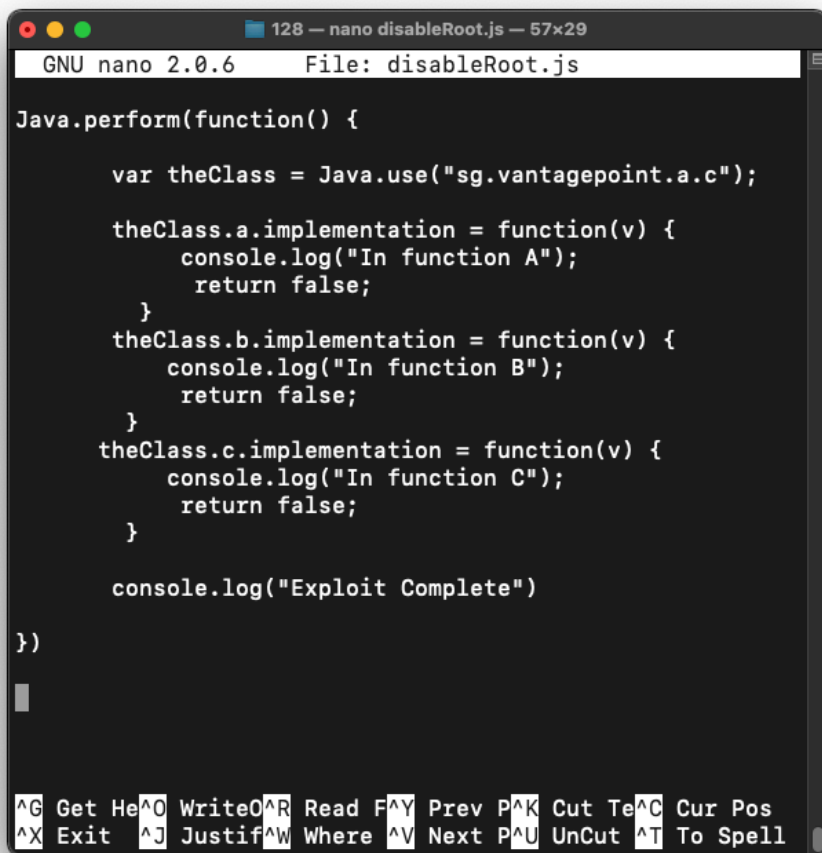
In a Terminal, execute this command:

```
nano disableRoot.js
```

Paste in this code, as shown below.

This code will override the a, b, and c functions and always return false.

```
Java.perform(function() {  
  
    var theClass = Java.use("sg.vantagepoint.a.c");  
  
    theClass.a.implementation = function(v) {  
        console.log("In function A");  
        return false;  
    }  
    theClass.b.implementation = function(v) {  
        console.log("In function B");  
        return false;  
    }  
    theClass.c.implementation = function(v) {  
        console.log("In function C");  
        return false;  
    }  
  
    console.log("Exploit Complete")  
  
})
```



```
GNU nano 2.0.6 File: disableRoot.js

Java.perform(function() {

    var theClass = Java.use("sg.vantagepoint.a.c");

    theClass.a.implementation = function(v) {
        console.log("In function A");
        return false;
    }
    theClass.b.implementation = function(v) {
        console.log("In function B");
        return false;
    }
    theClass.c.implementation = function(v) {
        console.log("In function C");
        return false;
    }

    console.log("Exploit Complete")

})


```

Type **Ctrl+x**, **y**, Enter to save the file.

Closing the App

On your Android device, click **OK** to close the app.

Exploiting the App

This command will launch the app in a frozen state, let the instrumentation occur, and then continue execution.

In a Terminal, execute this command:

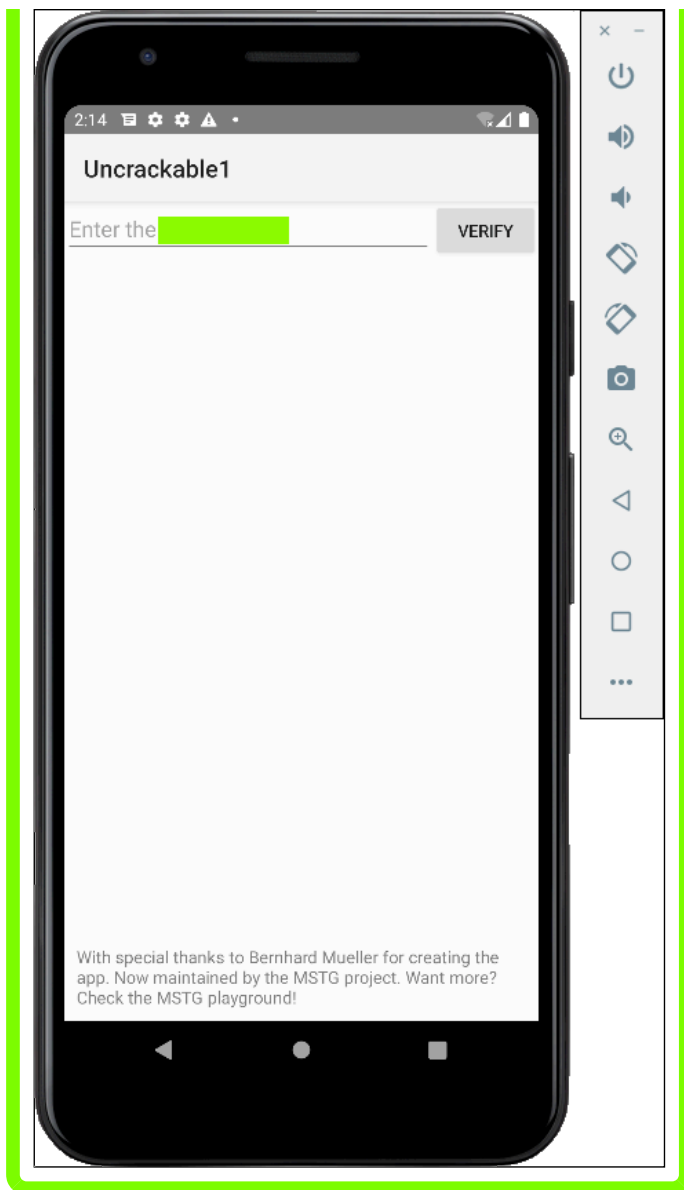
If you are using Frida version 16, omit the --no-pause switch

```
frida -U --no-pause -l disableRoot.js -f owasp.mstg.uncrackable1
```

M 513.1: Message

The app opens.

The flag is covered by a green rectangle in the image below.



Sources

[Android Hacking with FRIDA](#)

Posted 10-17-22

disableRoot.js filename fixed and note for Frida 16 added 10-17-22

Updated to Frida version 13 and Mac M1 host 3-20-24

"Externally managed environment" box added 3-28-24

Video added 4-18-24

Windows users note added 11-6-25